

Lithium-Ion Batteries with a Wide Temperature Range Operability Enabled by Highly Conductive sp^3 Boron-Based Single Ion Polymer Electrolytes

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We report excellent operability of Li-ion batteries equipped with selected sp^3 boron-based single-ion polymer electrolyte membranes (SIPEs) in the temperature range of 25–100 °C. The high performance of the batteries is attributed to the high ionic conductivity of the SIPEs, superior interfacial contacts between the SIPE membranes and the electrodes, and the sizes of the mesopores in the membranes being appropriate for ion transport. It is demonstrated that the use of or-

ganic solvents in the SIPE membranes does not lead to apparent molecular decomposition in the batteries even at temperatures as high as 100 °C, which underscores the exceptional performance of the SIPE membranes for safe operation of Li-ion batteries with excellent thermal and electrochemical stability. The intrinsic flexibility of the membranes enables this class of materials to be used in thin and flexible battery devices.

Introduction

Development of safe, cost-effective, and efficient Li secondary battery technologies with a wide temperature range operability has been a subject of intense current interest, especially in view of the recent safety incidences in the transportation sector.^[1,2] Although Li-ion battery technologies have long reached their maturity, there has been a constant market demand for better device performance and safety, longer cycle life, and higher energy densities. The improvements primarily rely on electrode materials with higher energy capacity and electrolytes with higher ionic conductivity and better thermal and electrochemical stability. Although the majority of research efforts have been on the development of electrode materials with better electrochemical performance and higher energy density, the choice of electrolyte has remained limited. The widely used conventional liquid electrolytes consisting of small-molecule Li salts dissolved in organic solvents display high ionic conductivity, but they suffer at the same time from potential safety issues, especially at elevated temperatures at which both the Li salts and the organic solvents may become chemically unstable.^[3–5] These Li salts include lithium tetrafluoroborate ($LiBF_4$),^[6–11] lithium bis(oxalato)borate ($LiBOB$),^[12,13] and lithium hexafluorophosphate ($LiPF_6$). Combustion of organic solvents owing to overheating, which could be caused by short circuiting between cathode and anode or operating at an elevated temperature, is deemed to be responsible for most of the safety-related incidences in battery devices.^[14] Today, it is still challenging to operate Li-ion batteries with liquid electrolytes above 60 °C.^[15] To overcome the safety problem, polymer electrolytes comprised of polymer composites and small-molecule inorganic Li salts have been developed, the advantages of which over liquid electrolytes are lower cost, longer longevity, and higher flexibility in size and shape.^[16–18] How-

ever, the mobility of anions and the low ion diffusivity in the polymer matrix give rise to serious concentration polarization, a low Li-ion transference number and low discharge capacity. Therefore, conventional polymer and gel polymer electrolytes are not well suited for applications that require a high power supply.^[19] In short, concentration polarization is an intrinsic property associated with dual-ion conducting electrolytes (small liquid-electrolyte molecules, conventional polymer or gel polymer electrolytes). Recently, single-ion conducting polymer electrolytes (SIPEs) have been developed as a new class of polymer electrolytes.^[20–22] These materials have been shown to enable charge and discharge to be conducted in Li-ion batteries more completely than in dual-ion electrolytes, partly because they reduce polarization because of immobile anions.^[21,23,24] In particular, several sp^3 boron-based single-ion-conducting polymer electrolytes have been found to display superior thermal stability and excellent electrochemical stability with a Li ion transference number

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close to unity.^[22,25] Nevertheless, to date, battery operability has only been demonstrated with a all-solid-state SIPE membrane at 80 °C,^[23] although several recent studies have shown that with appropriate treatment of the electrolyte membranes with small-molecule Li salts^[26] and/or on the triple-phase contact boundaries in the cathode^[27] Li-ion batteries with selected gel SIPE membranes can also be activated at room temperature with modest performance. Gel SIPEs, comprised of SIPE films and selected organic solvents, offer significantly better interfacial contact with the electrodes and, more importantly, substantially higher ionic conductivity than SIPEs without solvents and the conventional gel polymer electrolytes as ion transport can be facilitated through “lubrication” by the small organic molecules. Despite the advantages, the majority of Li-ion batteries equipped with gel SIPE membranes reported to date still work only at elevated temperatures. Furthermore, the use of organic solvents raises an important question as to whether the gel-SIPE-based batteries would still remain thermally and electrochemically stable.

The objective of the present study is to address the applicability of several gel-SIPE membranes in Li-ion batteries in a broad temperature range by demonstrating the operability of the batteries from 25 to 100 °C. In-depth analysis on the relationship between microstructures of the SIPE membranes and the battery performance was made through extensive structural characterizations and electrochemical measurements. Our results demonstrate that gel-SIPE-based batteries can exhibit excellent performance in the selected temperature range with unparalleled thermal and electrochemical stability compared with conventional polymer electrolytes and small-molecule Li salt liquid electrolytes. The inherently flexible nature of the SIPE membranes coupled with the high ionic conductivity and the broad temperature operability makes the Li-ion batteries highly promising for applications not only in thin, flexible electronics but also in devices that demand high power supplies such as electric vehicles.

In a previous publication, we reported the successful design and synthesis of three sp^3 boron-based single-ion-conducting polymer electrolytes with carboxyl linkers.^[28] These compounds exhibit thermal stability up to 570 °C and electrochemical stability up to 5.2 V (vs. Li^+/Li). However, the need for silylation of the carboxyl linkers for reaction with hexamethyldisilane under an absolutely anhydrous condition makes the synthesis difficult. In the present study, we demonstrate that a similar synthesis protocol can also be applied to the preparation of sp^3 boron-based single-ion-conducting polymer electrolytes through direct copolymerization of

phenol precursors with lithium tetramethanolatoborate [$LiB(OCH_3)_4$], bypassing the silylation process. Here, the compounds synthesized include lithium poly (hydroquinone borate) (LiPHB), lithium poly (hydroquinone borate) (LiPPB), and lithium poly (4,4'-biphenol borate) (LiPBB) as schematically shown in Figure 1. Characterization using a range of spectroscopic techniques reveals distinctively dif-

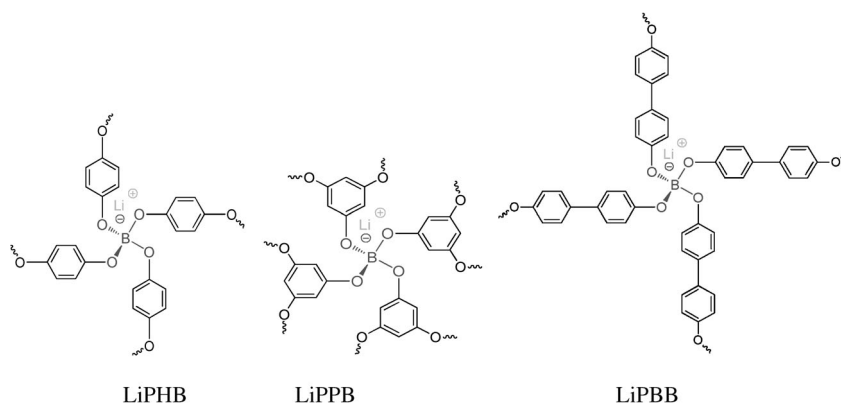


Figure 1. Structures of sp^3 boron-based SIPEs.

ferent morphologies and diffraction patterns in these materials. By analyzing the physicochemical properties and the structures of these compounds, we found that all batteries display excellent performance in a $Li||SIPE\ membranes||LiFePO_4$ cell at elevated temperatures. However, only batteries equipped with LiPHB and LiPPB membranes of an amorphous structure can be operated at room temperature, which is in contrast to the performance of batteries using the LiPBB membrane (no battery activities are observed under the same conditions despite the ordered structure of the polymer).

Results and Discussions

Morphology and microstructure characterization

Field emission (FE) SEM images and the X-ray diffraction patterns of LiPHB, LiPPB and LiPBB are shown in Figure 2. The broad peak in the range of 4° to 60° observed for LiPHB implies that its 3D framework is long-range disordered. The XRD spectrum of LiPPB exhibits a broad peak similar to that of LiPHB with several sharp diffraction features, indicating a slight improvement in the long-range order. In contrast, the XRD pattern of LiPBB consists of a set of very sharp diffraction peaks; thus, this material displays the best long-range order among the three compounds. The microstructural distinction of the three prepared SIPEs is consistent with the material morphologies obtained from SEM images. The morphology of LiPHB exhibits random packing of microspheres, whereas that of LiPPB displays packing of tile-shaped particles. In comparison, the morphology of LiPBB particles displays a cube-like shape with good crystallinity. The results indicate that there is a strong corre-

lation between the long-range order and the particle morphology.

The low crystallinity of LiPHB and LiPPB is likely due to the trapping of the unreacted precursors in the polymer matrix, resulting in long-range structural disorder. Indeed, the IR spectra depicted in Figure 9 exhibit a broad absorption band at 3440 cm^{-1} , corresponding to typical O–H stretching. To identify whether the absorption arises from water or from the precursor molecules, the samples were dried thoroughly at 100°C under reduced pressure (10^{-2} bar) for 24 h. However, the intensity of the absorption bands in all cases was not reduced, suggesting that the O–H stretching should be attributed to the residual phenol precursors trapped in the matrix upon polymerization instead of the absorbed water molecules. According to the IR spectrum of LiPBB, the phenol precursors were consumed more thoroughly during the polymerization process. Thus, the material exhibits better crystallinity.

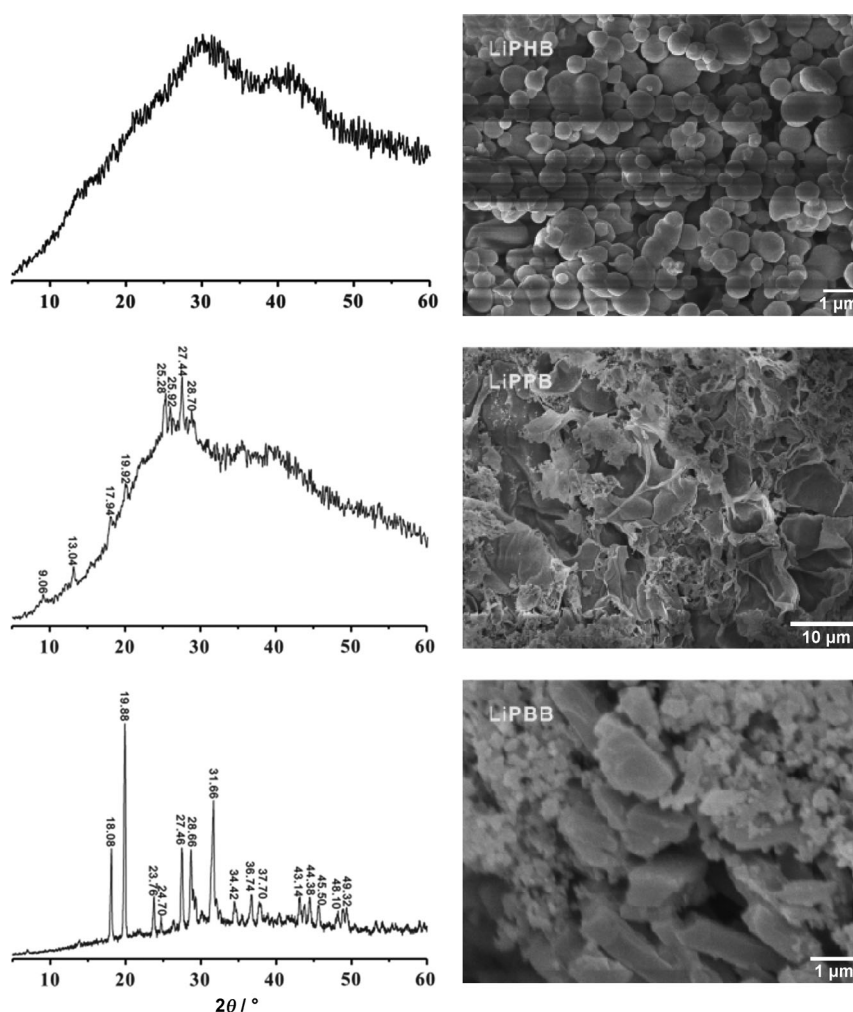


Figure 2. SEM images and XRD spectra of LiPHB, LiPPB, and LiPBB.

Thermal properties

The thermal stability of the synthesized materials was examined by thermal gravimetric analysis (TGA) under nitrogen. Thermal degradation of the compounds is shown in Figure 3. Compounds with the long-range disordered microstructures, LiPHB and LiPPB, exhibit inferior thermal stability with decomposition temperatures of 154 and 146°C , respectively. In contrast, the long-range-ordered structure, LiPBB, displays superior thermal stability up to 265°C . We further note that the TG curves under air show the same behavior as under nitrogen (Figure 4), indicating that the materials are stable in air. All three materials, especially the long-range-ordered compound (LiPBB), surpass the thermal stability requirement for Li based batteries ($<80^\circ\text{C}$)^[29] and may be suitable for high temperature applications.

Electrochemical stability

Linear sweep voltammetry (LSV) analysis was conducted to examine the electrochemical stability of the three compounds. Prior to the test, the SIPE membranes with the SEM images shown in Figure 5 were prepared through a solution

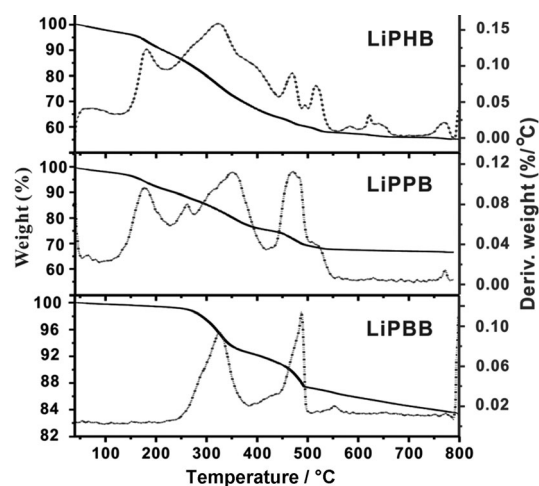


Figure 3. TGA curves of LiPHB, LiPPB, and LiPBB.

casting method. Clearly, the membranes are highly conformal and smooth, with mesopores and macropores serving as Li ion conduction channels. In particular, among the three membranes, the pore size of LiPHB is generally smaller and the pore size distribution in the membrane is more uniform,

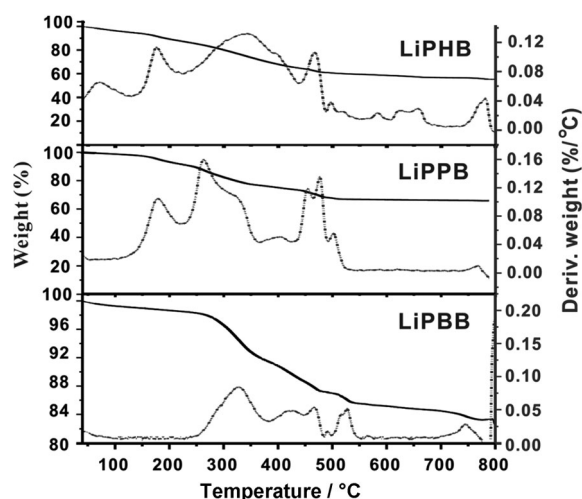


Figure 4. Thermo-oxidizing stability of LiPHB, LiPPB, and LiPBB.

whereas the pore size of the LiPBB membrane is significantly larger with large grain boundaries. An appropriate size of pores is important for battery performance. A too small size may block effective ion transport, whereas a too large size may not be beneficial either because Li ions must remain in close contact with the SIPE walls to maintain charge neutrality in ion conduction processes.

The SIPE membranes were used to construct Li||SIGE membranes||stainless steel cells. The current–voltage responses of the SIPEs in the cells at room temperature are shown in Figure 6. The LiPBB displays electrochemical stability up to 4.7 V, modestly higher than that of LiPHB (4.4 V) and LiPPB (4.4 V).

Li ion transference numbers and ionic conductivities

The steady-state current method was used to measure the Li ion transference number in a Li||blend polymer membranes||Li symmetric cell. The measured values for the parameters in Equation (1) and the derived Li^+ transference numbers are summarized in Table 1. The calculated Li^+ transference numbers of these materials are 0.93, 0.91, and 0.93 for LiPHB, LiPPB, and LiPBB, respectively. These values are substantially higher than that of LiBOB-based electrolytes (<0.6)^[30] and other electrolytes with small-molecule Li salts dissolved in organic solvents. It is well known that the transference number of Li^+ ions is sensitive to the mobility of the counter ions; the transference number increases as the

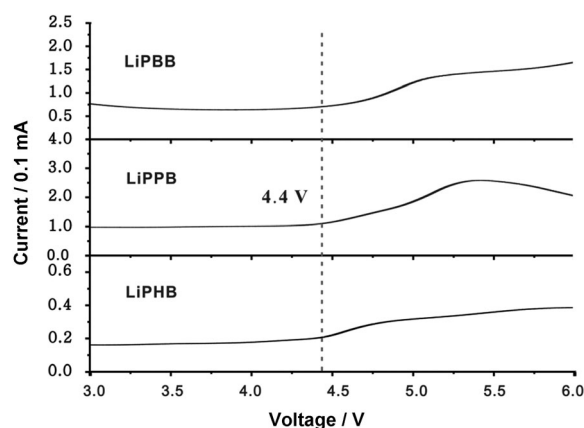


Figure 6. Electrochemical stability of LiPHB, LiPPB, and LiPBB.

Table 1. Measured values for the corresponding calculated values of Li ion transference numbers (t_{Li^+}).

Electrolyte	I_0 [μA]	I_s [μA]	R_0 [Ω]	R_s [Ω]	t_{Li^+}
LiPHB	0.0196	0.0183	9.3144	8.4006	0.9316
LiPPB	0.3214	0.2924	27.9300	28.1700	0.9097
LiPBB	1.3110	1.2180	12.1380	11.8040	0.9289

anions become bulkier.^[23] The high Li^+ transference numbers of the single-ion-conducting polymer electrolytes arise from the polyanionic structures remaining immobile in charge–discharge processes,^[21] which indeed underlines the unique advantage of SIPE materials.

The ionic conductivity of the prepared SIPE membranes in stainless steel (SS)||SIPE membranes||stainless steels (SS) cells was measured using electrochemical impedance spectroscopy (EIS). The obtained Nyquist plot includes two elements: a suppressed semicircle in the high-frequency regime corresponding to bulk resistance (R_b) and a straight line in the low-frequency regime associated with the interfacial resistance (R_i) controlled by diffusion of the charged particles. The ionic conductivity σ can be derived from the equation $\sigma = L/(RA)$, where L , A , and R represent the membrane thickness, the surface area, and the bulk resistance of the composite membrane, respectively. Figure 7 shows the Arrhenius and the Nyquist plots of PVdF-HFP/LiPHB, PVdF-HFP/LiPPB, and PVdF-HFP/LiPBB SIPE membranes. These membranes all exhibit high ionic conductivities above

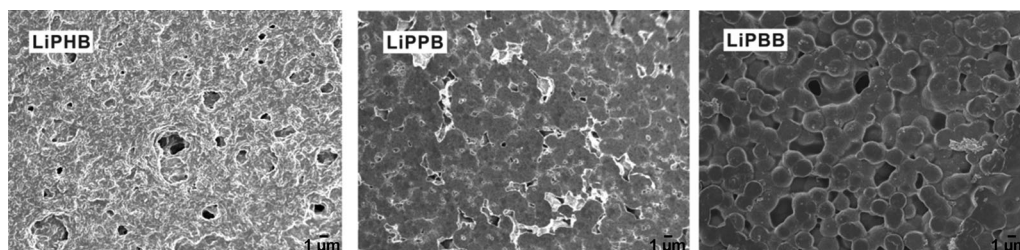


Figure 5. SEM images of the PVdFHFP/LiPHB, PVdFHFP/LiPPB, and PVdFHFP/LiPBB membranes.

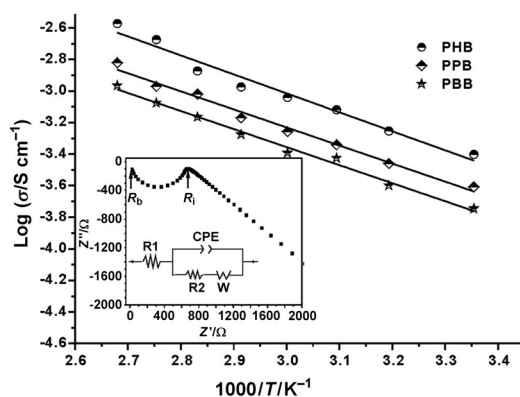


Figure 7. Arrhenius and Nyquist plots of the prepared PVdF-HFP/LiPHB, PVdF-HFP/LiPPB, and PVdF-HFP/LiPBB SIPE membranes.

$10^{-4} \text{ S cm}^{-1}$ at room temperature and even higher than $10^{-3} \text{ S cm}^{-1}$ at 100°C owing to the strong electron-withdrawing capability of the oxygen atoms and the significant p - π conjugation between the oxygen atoms and fused benzene rings as discussed above. The PVdF-HFP/LiPHB composite membrane displays the highest ionic conductivity of $2.50 \times 10^{-4} \text{ S cm}^{-1}$ at 25°C and $2.68 \times 10^{-3} \text{ S cm}^{-1}$ at 100°C , respectively, consistent with the smaller mesopores in the membrane. In comparison, the PVdF-HFP/LiPBB composite membrane shows the lowest ionic conductivity of $1.76 \times 10^{-4} \text{ S cm}^{-1}$ at 25°C and $1.08 \times 10^{-3} \text{ S cm}^{-1}$ at 100°C . This is likely due to the large macropores in the structure and lower Li content in LiPBB arising from the smaller molecular size of the hydroquinone precursor. The temperature dependency of the ionic conductivity of the SIPE membranes follows the typical Arrhenius equation over the temperature range of 25 – 100°C .

Battery performance

The membranes of the three electrolytes were individually assembled in coin cells with olivine (LiFePO_4) nanocrystals as the cathode and a Li foil as the anode for battery performance tests at different rates. Remarkably, batteries equipped with LiPHB and LiPPB electrolyte membranes display good performance at room temperature, as depicted in Figure 8a and b. The unusually active battery activity at room temperature is in sharp contrast to the performance of most of the SIPE-based batteries reported to date that display only high-temperature operability.^[23,31] The remarkable room-temperature

performance is attributed to the high ionic conductivity of these membranes and the low interfacial resistance between membrane and electrodes (Figure 8d). It is important to note that these two batteries also display excellent performance at 60°C with an average discharge capacity of 120 and 102 mAh g^{-1} at the high discharge rate of 1.0C and, most notably, at 100°C with an average discharge capacity of 102 and 90 mAh g^{-1} at 1.0C , respectively. The battery performance of the LiPHB and LiPPB membranes is particularly remarkable because the results suggest that batteries with the gel-SIPE membranes are fully operable in a broad temperature range. Clearly, including organic solvents in the SIPE matrix would not result in thermal and electrochemical instability at temperatures as high as 100°C , which is in sharp contrast to conventional liquid electrolytes and polymer electrolytes.^[3,32]

Unlike the LiPHB and LiPPB membrane-based batteries, the batteries equipped with the LiPBB electrolyte membrane could not be operated at room temperature. Good battery performance was observed only at an elevated temperature. As shown in Figure 8c, at 1.0C , the average discharge capacity was 122 mAh g^{-1} at 60°C and 120 mAh g^{-1} at 100°C . The poor room temperature performance of the LiPBB batteries is largely attributed to the relatively low ionic conductivity of the electrolyte and the high interfacial resistance compared to the batteries with the LiPHB and LiPPB membranes (Figure 8d).

Despite the good performance of the batteries with the SIPE membranes, especially at elevated temperatures, further improvements in ionic conductivity and interfacial resistance are still required to enhance the device performance

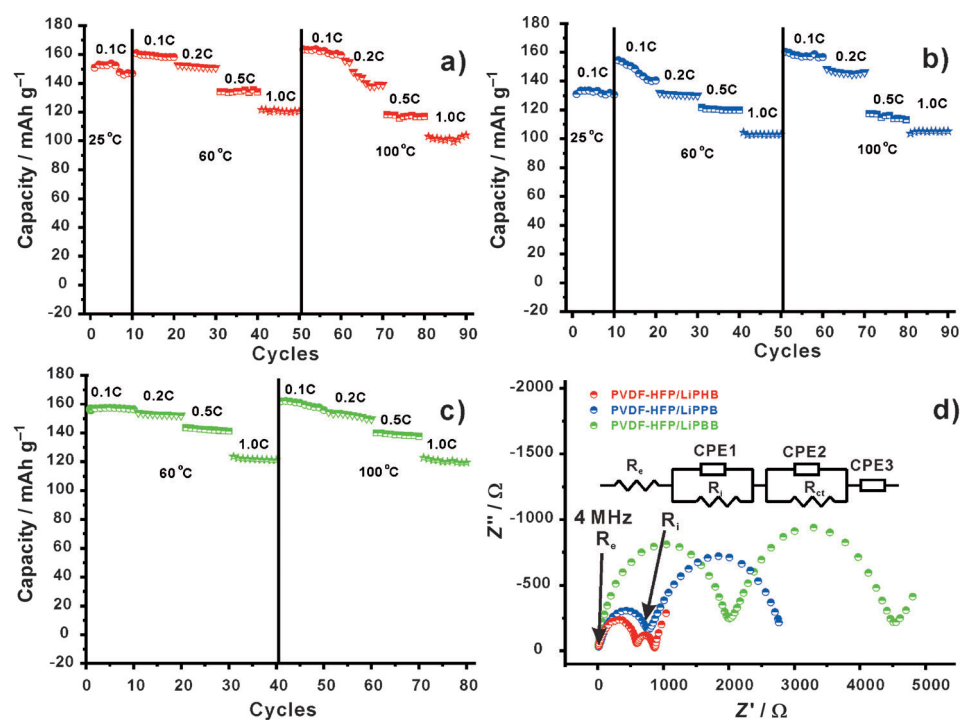


Figure 8. Cycling performances of the prototype $\text{Li} \parallel \text{SIPEs} \parallel \text{LiFePO}_4$ cells of a) LiPHB, b) LiPPB, and c) LiPBB and d) comparison of impedance plots at room temperature.

at room temperature. Incremental changes in Li content, reduction in thickness of the membranes, and generation of porosity with an appropriate pore size may be the steps to achieve this objective.

Conclusions

We have demonstrated for the first time the operability of Li-ion batteries equipped with selected gel-based single-ion polymer electrolyte (gel-SIPE) membranes in the temperature range of 25–100 °C. The high performance of the batteries arises from the high ionic conductivity of the SIPEs, the superior interfacial contacts between the SIPE membranes and the electrodes, and the appropriate mesopore sizes of the membranes for ion conduction. We have shown that the use of organic solvents, widely utilized in conventional liquid electrolytes and gel-polymer electrolytes of small-molecule Li salts, in SIPE membranes does not lead to apparent chemical decomposition in the batteries even at a temperature as high as 100 °C, which underlines the exceptional performance of the SIPE membranes for safe operation of Li-ion batteries with excellent thermal and electrochemical stability.

We have developed a general protocol to synthesize sp^3 boron-based single-ion conducting polymer electrolytes through direct copolymerization between lithium tetramethanolatoborate [$LiB(OCH_3)_4$] and phenol precursors, which is advantageous in comparison to carboxylic-acid-based precursors for which silylation is a necessary step to facilitate polymerization.^[23] Three prototypical polymeric ionic solids were obtained, and their membranes were fabricated by blending the polymers with PVdF-HFP. Subsequently, a variety of characterization techniques, including FTIR, ^{11}B NMR spectroscopy, gel permeation chromatography (GPC), SEM, and XRD, were utilized to determine the structures and properties of these membranes. It was found that the prepared SIPE membranes all exhibit high thermal stability up to 265 °C, high oxidative potential of approximately 4.5 V, and a high Li ion transference number of greater than 0.9. The measured ionic conductivity is essentially on the order of $10^{-3} \text{ S cm}^{-1}$ at 100 °C, comparable to the recently developed sp^3 boron-based SIPEs using carboxylic-acid-based precursors. The SEM images and the X-ray diffraction patterns indicate that these membranes also display various degrees of structural disordering arising from the extent of phenol precursors trapped in the polymer matrices. The trapped precursors in the polymer matrix give rise to an amorphous structure. Although the LiPBB-based batteries exhibit no activity at room temperature and stable performance can only be observed at elevated temperatures, the batteries constructed by using LiPHB and LiPPB display good stable performance at both room temperature and elevated temperatures up to 100 °C. The superior performance of the batteries at elevated temperatures underlines the importance of continuous improvement of ionic conductivity and interfacial resistance in SIPE membranes. The use of SIPE membranes in Li-ion batteries with a wide temperature range is expected to signifi-

cantly improve the device safety and to considerably broaden the applicability of the batteries.

Experimental Section

Materials

Hydroquinone ($\geq 99\%$), phloroglucinol ($\geq 99\%$), 4,4'-dihydroxy-biphenyl (97%), hexamethyldisilazane, dichloroethane, trimethyl borate, and poly(vinylidene fluoride-hexafluoropropylene) (PVdF-HFP), were purchased from Sigma-Aldrich. 1,2-Dichloroethane and acetonitrile were dried over phosphorus(V) oxide, methanol was dried over magnesium/iodine. Hydroquinone, phloroglucinol, and 4,4'-dihydroxy-biphenyl were dried at 80 °C under reduced pressure overnight.

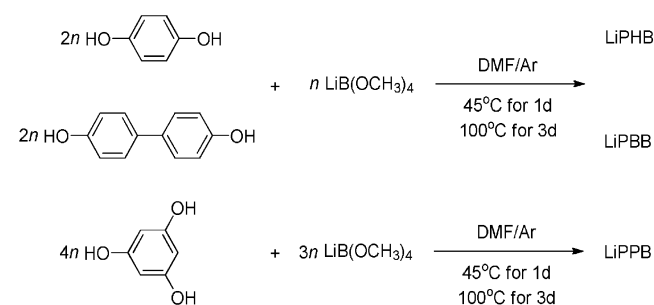
Synthesis

Synthesis of lithium tetramethanolatoborate

Lithium tetramethanolatoborate [$LiB(OCH_3)_4$] was synthesized and purified according to the procedure described by Barthel et al.^[33] Li (0.347 g) was added into a Pyrex flask in a glove box; the Pyrex flask was then transferred from the glove box to a fume hood and immediately connected to an argon flow line. The flask was kept in an ice bath as the reaction is highly exothermic. Methanol (11 mL) was added slowly to the flask, and subsequently another portion of methanol (14 mL) was added. The mixture was heated to 60 °C to ensure that the solution is homogeneous. Boric acid trimethyl ester (5.2 g; 5.7 mL) was then slowly dropped into the solution. Finally, the reaction was kept for another 24 h at room temperature for crystallization, filtration, and desiccation, yielding colorless nanoparticles with a yield of 92%. FTIR (KBr, Avatar-37, 500–4000 cm^{-1}): $\tilde{\nu}=3422$ (s; ν_{C-H}), 1385 (s; ν_{B-O}), 1202 (s; ν_{C-O}), 990.5 cm^{-1} (s; ν_{C-O}) strong peak; 1H NMR (300 MHz, $[D_6]DMSO$): $\delta=3.16$ ppm (s, 12 H, CH_3); ^{13}C NMR ($[D_6]DMSO$): $\delta=48.95$ ppm (CH_3); ^{11}B NMR (500 MHz, $[D_6]DMSO$): $\delta=3.37$ ppm (s, sp^3 boron); ESI-MS $[(M-Li)^-]$ m/z 135.0 (the molecular weight of $C_4H_{12}O_4B$).

Synthesis of SIPEs

The sp^3 boron-based SIPEs were synthesized by directly reacting lithium tetramethanolatoborate [$LiB(OCH_3)_4$] with the phenol linkers under mild conditions as shown in Scheme 1. The stoichiometric lithium tetramethanolatoborate and phenol precursors were first dissolved in DMF (Baker's HPLC grade, dried); the reaction was then carried out at 45 °C for 1 d and subsequently at 70 °C for another 3 d. After that, the solution was



Scheme 1. Procedures for the synthesis of the SIPEs.

cooled down followed by addition of dried acetonitrile to separate the products from the solution. The raw products were collected by filtration and then purified through recrystallization with DMF followed by further purification through washing continuously with acetonitrile. After drying at 120 °C under vacuum for 24 h, the products were characterized by FTIR, ^{11}B NMR spectroscopy, and GPC. The characteristic absorption in the range 1000–1400 cm^{-1} confirmed the formation of the B–O bond (Figure 9). The ^{11}B NMR spectra exhibited peaks in the range of 0–10 ppm, demonstrating that all boron atoms adopted a sp^3 configuration (Figure 10). Molecular weight of the prepared materials, shown in Table 2, was obtained from the GPC analysis with water as the solvent. The number-average molecular weight (M_n) of the prepared SIPEs was in the range of 16400–27800 and polydispersity index (M_w/M_n) was between 1.32 and 2.13, suggesting a relatively narrow distribution of the molecular weights.

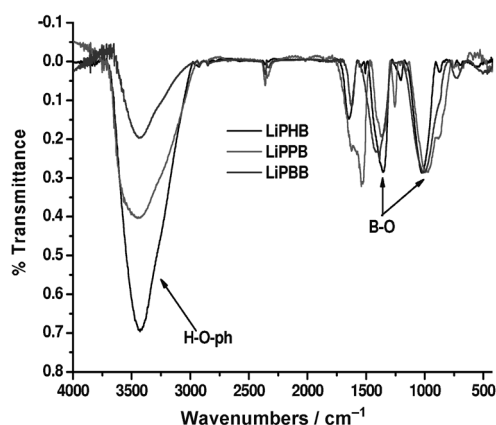


Figure 9. FTIR spectra of the SIPEs.

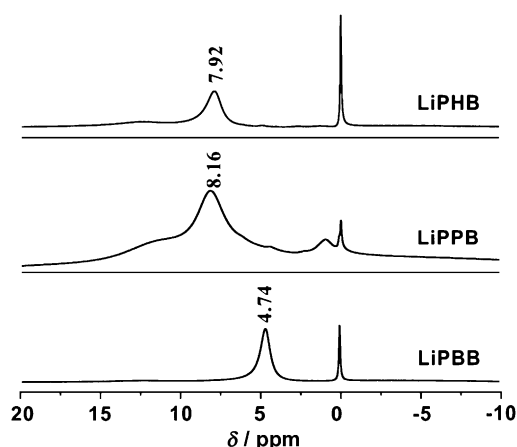


Figure 10. ^{11}B NMR spectra of the SIPEs.

Table 2. Molecular weights of the SIPEs by GPC analysis.			
Polymer	M_n [$\times 10^4 \text{ g mol}^{-1}$]	M_w [$\times 10^4 \text{ g mol}^{-1}$]	M_w/M_n
LiPHB	2.78	3.68	1.32
LiPPB	1.85	2.50	1.35
LiPBB	1.64	3.48	2.13

Preparation of single ion conducting membranes

Homogeneous solutions were first obtained by dissolving PVdF-HFP and the prepared SIPEs at a weight ratio of 3:1 into DMF solvent under 80 °C. The above solutions were subsequently cast onto a glass plate, and the solvent was evaporated slowly at 80 °C. The remaining DMF was fully removed under vacuum at 80 °C for two days, and the dried SIPE membranes were transferred into a glove box and immersed in EC/PC (v/v , 1:1) for further characterizations. The thickness of the obtained polymer electrolyte membranes was around 30 μm .

Scanning electron microscopy

The surface morphology of the PVdF-HFP/SIPE membranes was investigated by using a JEOL JSM-6701F field emission scanning electron microscope. Test specimens were sputter coated with platinum before recording the image.

X-ray diffraction spectroscopy

The X-ray diffraction patterns of the samples were obtained by using a computer-controlled X'pert PRO-PANalytical diffractometer with $\text{CuK}\alpha$ radiation (wavelength, $\lambda = 0.1541 \text{ nm}$) as the source at 40 kV with a scan range between 1.4° and 60°.

Thermal analysis

Thermal analysis of the PVdF-HFP/SIGE membranes was conducted by applying a 10.00 °C min^{-1} ramp up to 800 °C using a SDT TA Instruments 2960 Simultaneous DTA-TGA under both nitrogen and air.

Electrochemical characterization of the PVdF-HFP/SIGE membranes.

The electrochemical stability of the PVdF-HFP/SIGE membranes was evaluated by performing a linear sweep voltammetry (LSV) experiment at a scan rate of 0.2 mVs^{-1} in a sandwich cell using two SS electrodes both, as working electrode and counter electrode. The ionic conductivity of the PVdF-HFP/SIGE membrane was determined by EIS using a potentiostat-galvanostat AUTOLAB model PGSTAT12/30/302 over a frequency range of 0.1 Hz to 4 MHz with the oscillating voltage of 5 mV using the above-described cell. The Li ion transference number, t^+ , was calculated according to the following formula described by Evans.^[34]

$$t^+ = \frac{I_s(\Delta V - I_0 R_0)}{I_0(\Delta V - I_s R_s)} \quad (1)$$

where ΔV is the potential applied across the cell, I_0 and I_s are the initial and steady-state currents, and R_0 and R_s are the initial and steady-state resistances of the passivation layers on the Li electrode.

Battery performance tests were carried out by sandwiching the PVdF-HFP/SIPE membranes between a Li metal foil and a LiFePO_4 electrode in a coin cell (2025-type). The LiFePO_4 electrode was prepared by casting a homogeneous mixture of LiFePO_4 , carbon black, PVdF, and bis(4-carboxyl benzene sulfonyl) imide (CBSI)^[35] with the mass ratio of 0.75/0.1/0.1/0.05 ($w/w/w/w$) onto an aluminum foil. The cell was charged and dis-

charged at different rates in a voltage range of 3.8–2.5 V using an Arbin BT 2000.

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Keywords: batteries • ionic conductivity • lithium • polymer electrolytes • temperature

- [1] B. Scrosati, J. Garche, *J. Power Sources* **2010**, *195*, 2419.
- [2] J. M. Tarascon, M. Armand, *Nature* **2001**, *414*, 359.
- [3] S. E. Sloop, J. K. Pugh, S. Wang, J. B. Kerr, K. Kinoshita, *Electrochem. Solid-State Lett.* **2001**, *4*, A42.
- [4] S.-Y. Li, P.-H. Ma, X.-L. Cui, Q.-D. Ren, F.-Q. Li, *J. Chem. Sci.* **2008**, *120*, 289.
- [5] L. J. Krause, W. Lamanna, J. Summerfield, M. Engle, G. Korba, R. Loch, R. Atanasoski, *J. Power Sources* **1997**, *68*, 320.
- [6] A. B. McEwen, H. L. Ngo, K. LeCompte, J. L. Goldman, *J. Electrochem. Soc.* **1999**, *146*, 1687.
- [7] R. Dupon, B. L. Papke, M. A. Ratner, D. F. Shriver, *J. Electrochem. Soc.* **1984**, *131*, 586.
- [8] C. Li, T. Imae, *Macromolecules* **2004**, *37*, 2411.
- [9] M. S. Su'ait, S. A. M. Noor, A. Ahmad, H. Hamzah, M. Y. A. Rahman, *J. Solid State Electrochem.* **2012**, *16*, 2275.
- [10] S. D. Arnó, S. Lucas, A. Shariati, C. J. Peters, *J. Solution Chem.* **2012**, *41*, 2173.
- [11] M. Ue, K. Ida, S. Mori, *J. Electrochem. Soc.* **1994**, *141*, 2989.
- [12] U. Lishka, U. Wietelmann, M. Wegner (Metallgesellschaft Ag), DE19829030C1, **1999**.
- [13] W. Xu, C. A. Angell, *Electrochem. Solid-State Lett.* **2001**, *4*, E1.
- [14] K. O. Christe, D. A. Dixon, D. McLemore, W. W. Wilson, J. A. Sheehy, J. A. Boatz, *J. Fluorine Chem. Commun.* **2000**, *101*, 151.
- [15] K. Tikhonov, V. R. Kock, Covalent Associates, Inc., Woburn, MA.
- [16] G. S. MacGlashan, Y. G. Andreev, P. G. Bruce, *Nature* **1999**, *398*, 792.
- [17] X. H. Flora, M. Ulaganathan, S. Rajendran, *Int. J. Electrochem. Sci.* **2012**, *7*, 7451.
- [18] H. W. Chen, T. P. Lin, F. C. Chang, *Polymer* **2002**, *43*, 5281.
- [19] J. N. Chazalviel, *Phys. Rev. A* **1990**, *42*, 7355.
- [20] R. Mezziane, J.-P. Bonnet, M. Courty, K. Djellab, M. Armand, *Electrochim. Acta* **2011**, *57*, 14.
- [21] S. W. Feng, D. Y. Shi, F. Liu, L. P. Zheng, J. Nie, W. F. Feng, X. J. Huang, M. Armand, Z. B. Zhou, *Electrochim. Acta* **2013**, *93*, 254.
- [22] Y. S. Zhu, X. J. Wang, Y. Y. Hou, X. W. Gao, L. L. Liu, Y. P. Wu, M. Shimizu, *Electrochim. Acta* **2013**, *87*, 113.
- [23] R. Bouchet, S. Maria, R. Mezziane, A. Aboulaich, L. Lienafa, J.-P. Bonnet, T. N. T. Phan, D. Bertin, D. Gigmes, D. Devaux, R. Denoyel, M. Armand, *Nat. Mater.* **2013**, *12*, 452.
- [24] K. Matsumoto, T. Endo, *J. Polym. Sci. Part A* **2011**, *49*, 1874.
- [25] X. G. Sun, J. B. Kerr, *Macromolecules* **2006**, *39*, 362.
- [26] Y. Zhu, S. Xiao, Y. Shi, Y. Yang, Y. Wu, *J. Mater. Chem. A* **2013**, *1*, 7790.
- [27] W. Cai, Y. Zhang, J. Li, Y. Sun, H. Cheng, *ChemSusChem* **2014**, *7*, 1063.
- [28] Y. Zhang, G. Xu, Y. Sun, B. Han, T. B. W. T, Z. Chen, R. Rohan, H. Cheng, *RSC Adv.* **2013**, *3*, 14934.
- [29] J. H. Shin, E. J. Cairns, *J. Electrochem. Soc.* **2008**, *155*, A368.
- [30] W. Xu, A. J. Shusterman, M. Videa, V. Velikov, R. Marzke, C. A. Angell, *J. Electrochem. Soc.* **2003**, *150*, E74.
- [31] X. Wang, Z. Liu, C. Zhang, Q. Kong, J. Yao, P. Han, W. Jiang, H. Xu, G. Cui, *Electrochim. Acta* **2013**, *92*, 132.
- [32] D. Aurbach, Y. Talyosef, B. Markovsky, E. Markevich, E. Zinigrad, L. Asraf, J. S. Gnanaraj, H.-J. Kim, *Electrochim. Acta* **2004**, *50*, 247.
- [33] J. Barthel, R. Buestrich, E. Carl, H. J. Gores, *J. Electrochem. Soc.* **1996**, *143*, 3572.
- [34] J. Evans, C. A. Vincent, P. G. Bruce, *Polymer* **1987**, *28*, 2324.
- [35] Y. F. Zhang, J. W. Y. Ting, R. Rohan, W. W. Cai, J. Li, G. D. Xu, Z. X. Chen, A. Lin, H. S. Cheng, *J. Mater. Sci.* **2014**, *49*, 3442.

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